

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	Sanjay kumar soy
Register Number	192311151

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I sanjay kumar soy(192311151) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

sanjay kumar soy

Signature of the student:

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Answer

1.A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workload

Cloud Architecture

The videos can be stored in cloud storage, while a Content Delivery Network (CDN) keeps copies of popular videos at locations closer to users. This reduces loading time and buffering. A load balancer distributes user requests across multiple application servers so that no single server becomes overloaded. The application servers handle user login, video streaming, and recommendations, while a cloud database stores user information and viewing history. The entire system is monitored continuously to ensure smooth performance.

Role of Load Balancing and Auto-Scaling

- Load balancing ensures that incoming traffic is shared evenly among all available servers. If one server becomes busy or fails, requests are automatically redirected to another server, improving reliability.
- Auto-scaling automatically increases the number of servers when many users are streaming videos and reduces them when traffic decreases. This helps maintain smooth video playback without buffering while avoiding unnecessary resource usage.

Cost Optimization Strategies

The company can reduce costs by using auto-scaling so that it only pays for the resources it actually needs. Frequently watched videos can be cached in the CDN, reducing bandwidth and server load. Less popular videos can be moved to low-cost cloud storage. Regular monitoring of resource usage also helps identify and remove unused resources, further lowering operational costs.



2.A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Solution Using Container Orchestration and Multi-Cloud

The startup can package its applications as Docker containers and manage them using Kubernetes. Kubernetes provides a common platform to deploy and manage containers across multiple cloud providers such as AWS, Azure, and Google Cloud. This ensures that the application behaves the same way regardless of the cloud environment and simplifies deployment and management.

Service Discovery and Scaling

Service discovery allows containers to automatically find and communicate with each other without manually configuring network addresses. Kubernetes provides built-in service discovery using DNS and services.

For scaling, Kubernetes automatically increases or decreases the number of container instances based on CPU usage, memory usage, or user traffic. This helps the application handle sudden increases in demand while maintaining good performance.

Benefits and Risks of Multi-Cloud Deployment

Benefits:

- Improves availability by avoiding dependence on a single cloud provider.
- Provides better disaster recovery if one cloud provider experiences an outage.
- Allows the organization to choose the best services and pricing from different providers.
- Increases flexibility and reduces vendor lock-in.

Risks:

- Managing multiple cloud platforms increases operational complexity.
- Security and compliance become more challenging across different environments.
- Data synchronization between cloud providers can be difficult.
- Monitoring and troubleshooting require additional tools and expertise.

3.An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures

a) Solution Using Backup, Replication, and Versioning

The platform should perform regular cloud backups of databases, user records, and course materials. Data replication should be enabled so that copies of the data are stored in different cloud regions or availability zones. Versioning should also be enabled, allowing previous versions of files to be restored if data is accidentally deleted or modified. This provides multiple layers of protection against data loss.

b) Achieving RTO and RPO

- Recovery Time Objective (RTO) is the maximum time required to restore the system after a failure. It can be reduced by keeping backup copies readily available and using automated recovery processes.
- Recovery Point Objective (RPO) is the maximum amount of data that can be lost. It can be minimized by taking frequent backups and continuously replicating data to another location.

Together, these measures ensure that the platform can recover quickly with minimal data loss.

c) Steps to Ensure Business Continuity

- Schedule regular automatic backups of all important data.
- Enable data replication across multiple regions or availability zones.
- Use versioning to recover deleted or modified files.
- Test backup and disaster recovery plans regularly.
- Continuously monitor the system and respond quickly to failures.

4.A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

A government agency can use a hybrid cloud architecture to meet security and compliance requirements. Sensitive data and critical applications should remain on-premises, while non-sensitive applications and public services can be hosted in the public cloud. A secure VPN or dedicated private connection should be used to enable safe communication between the two environments.

i) Hybrid Architecture

In this model, confidential citizen data and government records remain in the on-premises data center, while less sensitive applications such as public websites, email services, or online portals are deployed in the public cloud. A secure network connection ensures smooth and protected communication between both environments.

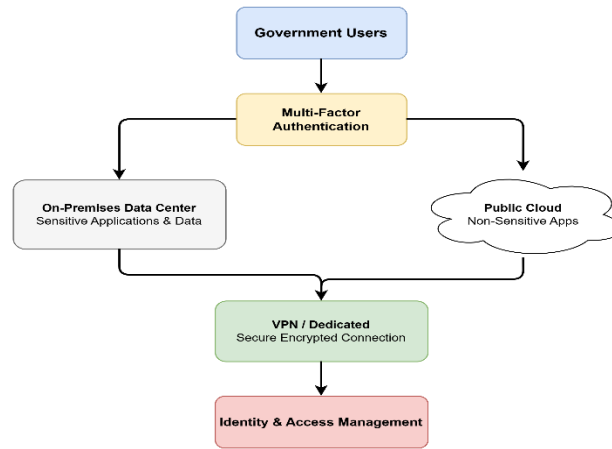
ii) Identity Management and Encryption

A centralized Identity and Access Management system should be used so that employees can securely access resources in both the on-premises and cloud environments using a single identity. Multi-Factor Authentication should be enabled to strengthen security.

Data should be encrypted both at rest and in transit. Encryption protects stored data from unauthorized access, while secure communication protocols such as TLS/SSL protect data as it travels between the on-premises infrastructure and the cloud.

iii) Challenges in Managing a Hybrid Cloud

- Managing security policies across different environments can be complex.
- Ensuring compliance with government regulations requires continuous monitoring.
- Integrating on-premises systems with cloud services may be challenging.
- Maintaining consistent performance and network connectivity between environments requires careful planning.
- Monitoring and managing resources across multiple platforms can increase administrative effort.



5.A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Multi-Region Deployment Strategy

The application should be deployed in two or more cloud regions. Each region should have its own application servers, load balancers, and databases. A global load balancer can direct users to the nearest or healthiest region. If one region becomes unavailable, user traffic is automatically redirected to another region without affecting the service.

Data Replication and Failover

Data should be replicated continuously between the cloud regions to keep all copies synchronized. This ensures that the latest data is available in every region. An automatic failover mechanism should be configured to detect failures and switch users to the backup region immediately. This reduces downtime and prevents data loss during outages.

Monitoring and Alerting

The system should be monitored continuously using cloud monitoring tools to track server health, CPU usage, memory, network performance, and application availability. Alerts should be configured to notify administrators immediately if any failures or unusual activities occur. Regular health checks and automated monitoring help identify problems early and ensure quick recovery.

